



TERRASAFE:

Revolutionizing Portable Navigation for GPS-Denied Environments

Standard navigation tools like GPS/GNSS are ineffective in many environments, including dense forests, caves, subways, and areas with electronic interference. This poses a significant risk for professionals and adventurers such as first responders, soldiers, hikers, and urban explorers, who need to maintain situational awareness for safe navigation and return. The critical challenge is the lack of a self-contained, portable device that can provide real-time position tracking and reverse navigation without relying on external signals.

Team Behind TERRASAFE

Our dedicated team brings together expertise in embedded systems, sensor technology, and user interface design to create robust navigation solutions.



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TERRASAFE: Core Hardware Components

The TERRASAFE device is engineered with a robust selection of components to ensure reliable performance in challenging environments.

Microcontroller

ESP32: The brain of the device, handling all data processing and system operations.

Environmental Sensor

BMP180: Measures barometric pressure and temperature for altitude tracking and environmental awareness.

Motion & Orientation Sensors

- MPU6050 (Accelerometer & Gyroscope)
- QMC5883L (Magnetometer)

These provide crucial data for tracking movement and direction.

Display & Power

- OLED Display (SSD1306): Compact screen for real-time data visualization.
- Push Button: Simple user input for mode selection.
- Battery: Powers the device for extended operation.

The TERRASAFE Solution: Dead Reckoning & Layered Architecture

TERRASAFE navigates by tracking direction and distance from a starting point, powered by a sophisticated layered system.

Dead Reckoning: Navigating Without Signals

At its core, TERRASAFE uses dead reckoning to calculate your position by estimating direction and distance from a start point, making it ideal for GPS-denied environments (e.g., dense forests, underground structures). The device uses onboard sensors (accelerometer, gyroscope, magnetometer) to track movement, processing the raw data to filter noise and drift for an accurate, real-time position estimate. This allows users to maintain situational awareness in challenging conditions.

TERRASAFE's Robust Layered Architecture

The effectiveness of dead reckoning is amplified by TERRASAFE's carefully designed layered architecture, which ensures modularity, reliability, and efficient data flow.

Hardware Layer

This foundational layer comprises all the physical components essential for data collection and interaction. It includes the suite of sensors (accelerometer, gyroscope, magnetometer, barometric pressure sensor), the ESP32 microcontroller for processing, the OLED display for output, and the push button for user input. This layer is responsible for raw data acquisition and basic system operations.

Data Processing & Fusion Layer

Runs at 50Hz, performs sensor calibration (gyro bias removal), and fuses data using advanced algorithms like **Autocorrelation**, **Kalman Filter** (for position smoothing) and **Complementary Filter** (for accurate altitude: 70% barometric, 30% accelerometer).

Navigation Logic Layer

Utilizes fused data for movement interpretation: **Autocorrelation Pedometer Algorithm** for step counting, **Breadcrumb Trail System** for path recording and reverse routing, and **Waypoint System** for proximity alerts to saved points.

Presentation & Interface Layer

User interaction: Onboard OLED UI (Map, Dashboard, Reverse Navigation modes) and a **Remote Web Interface** via WiFi for detailed map viewing, waypoint management, and real-time sensor monitoring on external devices.

TERRASAFE Functionality & Intuitive User Interface

TERRASAFE offers comprehensive navigation features and flexible user interaction for diverse operational needs.

Core Functionalities

- **Real-time Dead Reckoning:** Tracks X, Y, Z position relative to start.
- **Enhanced Breadcrumb Navigation:** Provides turn-by-turn directions to retrace paths.
- **Smart Waypoint System:** Save up to 10 custom markers with proximity alerts.
- **Dynamic Mapping:** Real-time 2D path plot on OLED or web interface.
- **Environmental Data:** Real-time temperature and pressure readings.

User Interface



Onboard OLED Display

Multiple modes including Map Screen, 3D Dashboard, and Reverse Navigation for quick checks.



Remote Web Interface

Full-featured HTML5 map via local Wi-Fi, offering detailed viewing, waypoint management, and real-time sensor monitoring without internet.

TERRASAFE Project Outcomes

The TERRASAFE project successfully delivered a self-contained, portable navigation device for GPS-denied environments.

Functional Prototype

A working device that tracks relative position and provides reverse-path navigation.

Dual-Interface System

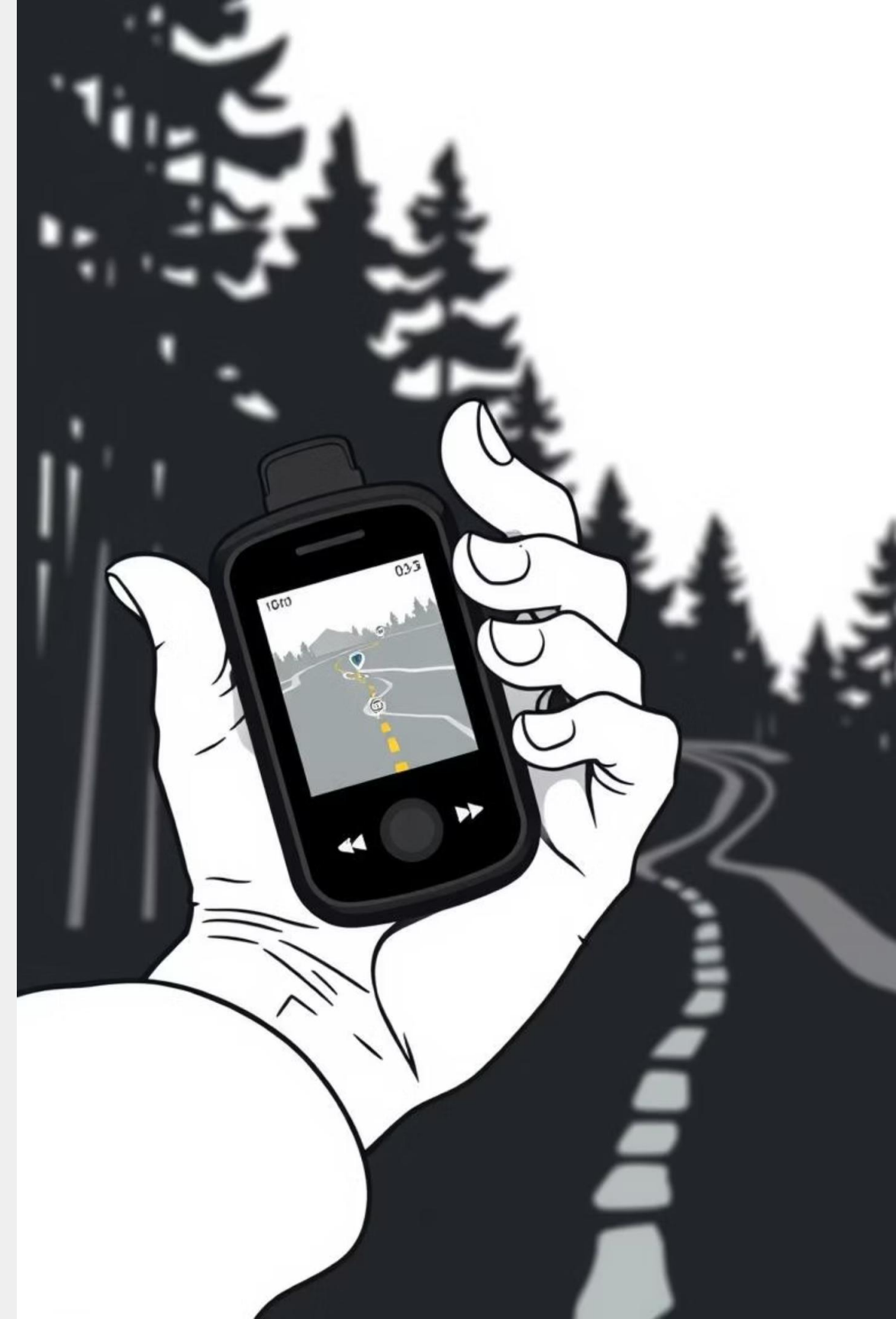
Seamless access to navigation data via compact OLED and detailed web map.

Dynamic Path Visualization

Real-time map generation without reliance on pre-loaded maps.

Identified Limitations

Acknowledged inherent dead reckoning challenges to guide future development.



Conclusion & Future Scope

TERRASAFE is a robust tool for off-grid safety, with a clear roadmap for continuous enhancement and broader application.

Conclusion

TERRASAFE demonstrates the power of integrating dead reckoning, sensor fusion, and breadcrumb navigation. Its adaptable architecture and user-centric design provide a strong foundation for a scalable and powerful navigation solution.

Future Scope & Next Steps

- **Machine Learning Pedometer:** Improve accuracy with dynamic stride length adjustment.
- **LoRa Mesh Networking:** Enable collaborative mapping for teams.
- **Enhanced Power Management:** Extend battery life for multi-day expeditions.